

Homework #1 for Geospatial Analysis & Remote Sensing (EDS223 - Fall 2026)

2026-10-10

```
# Load in Packages
library(tidyverse)
library(sf)
library(here)
library(tmap)

# Read in geodatabase of EJScreen data at the Census Block Group level
ej <- sf::st_read(here::here(
  "data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))

# Filter dataset to a state
louisiana <- ej |>
  dplyr::filter(ST_ABBREV == "LA")

# Filter dataset to a county
east_baton <- louisiana |>
  dplyr::filter(CNTY_NAME == "East Baton Rouge Parish")

# Find the average values for all variables within counties
ks_counties <- aggregate(louisiana, by = list(louisiana$CNTY_NAME),
  FUN = mean)

# Make the map output static
tmap_mode("plot")
```

```

# Map of the proportion of people of color in East Baton Rouge Parish

# Specifying dataframe from filtered data
tm_shape(east_baton) +

# Fill polygon shape by people of color variable while
# customizing specific attributes to enhance aesthetic
  tm_polygons(fill = "PEOPCOLORPCT",
    lwd = 0.1,
    col = "white",
    fill.scale = tm_scale(values = "brewer.blues"),
    fill.legend = tm_legend(title = "People of Color (proportion)",
      position = tm_pos_out("right", "center")))) +

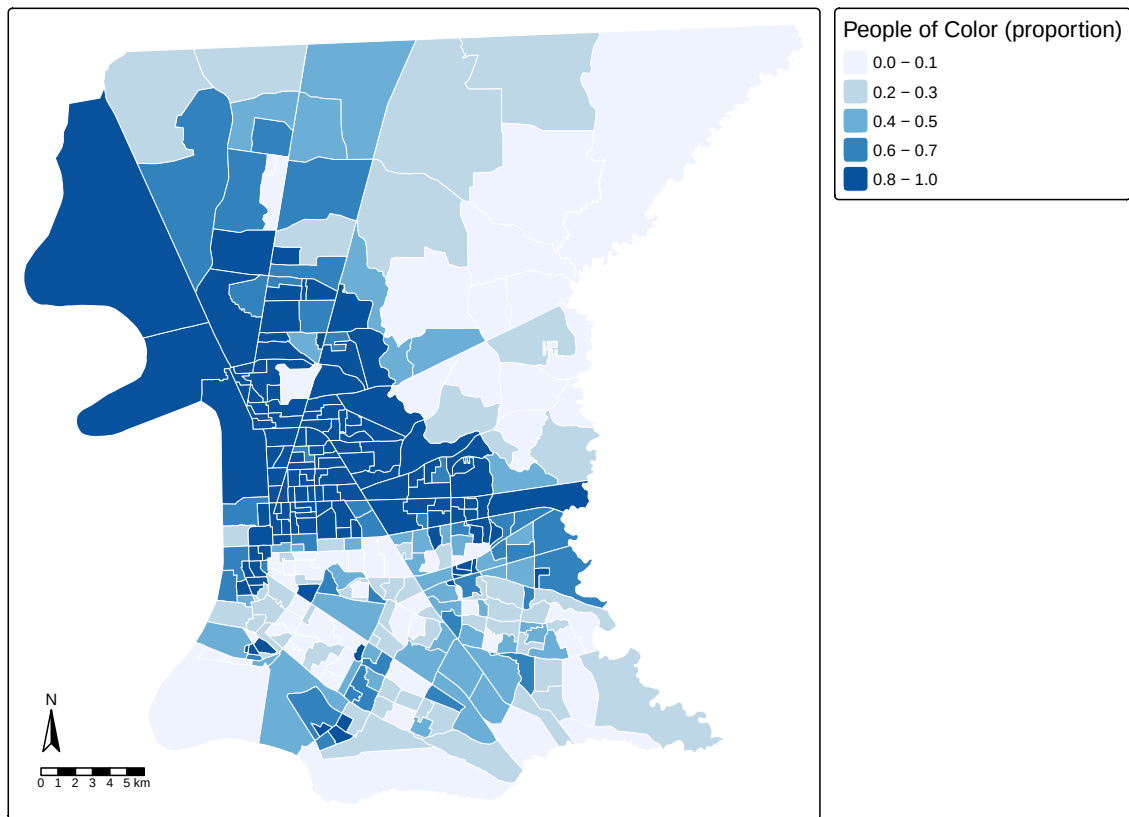
# Add a north arrow in the bottom left corner
  tm_compass(
    size = 2, text.size = 0.7, position = c("left", "bottom")) +

# Adding a scalebar in the bottom left corner
  tm_scalebar(
    width = 8, text.size = 0.5, position = c("left", "bottom")) +

# Add a title above the map, positioned to the left
  tm_title(
    "People of Color\nin East Baton Rouge Parish, LA", #
    size = 1.0,
    position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom"))

```

People of Color
in East Baton Rouge Parish, LA



```
# Map of the Proximity to RMP (chemical) facilities in East Baton Rouge Parish

# Specifying dataframe from filtered data
tm_shape(east_baton) +

# Fill polygon shape with proximity to RMP facility variable while
# customizing specific attributes to enhance aesthetic
tm_polygons(fill = "PRMP",
  lwd = 0.1,
  col = "black",
  fill.scale = tm_scale_intervals(
    style = "quantile",
    values = c("yellow", "darkred")),
```

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    fill.legend = tm_legend(title = "RMP Facility Proximity\n(facilities/km)",
    position = tm_pos_out("right", "center"))) +

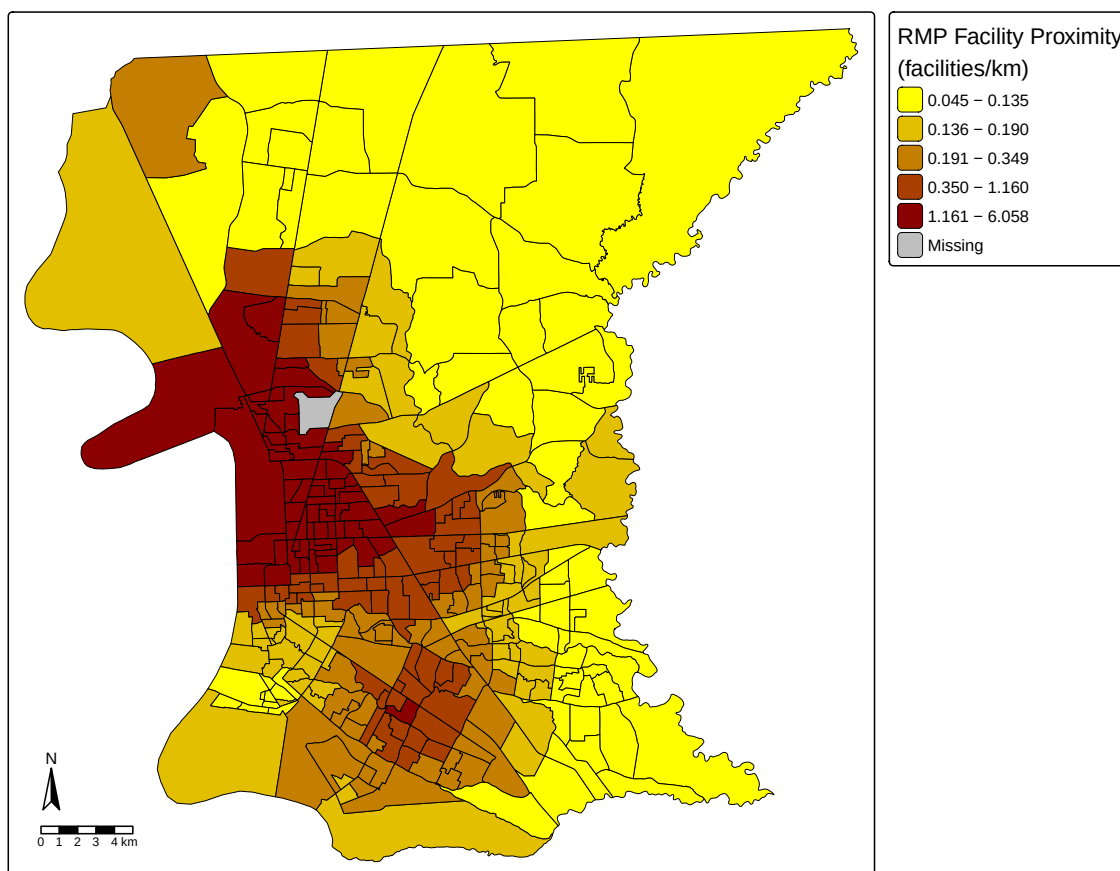
# Add a north arrow in the bottom left corner
tm_compass(
    size = 2, text.size = 0.7, position = c("left", "bottom")) +

# Adding a scalebar in the bottom left corner
tm_scalebar(
    width = 8, text.size = 0.5, position = c("left", "bottom")) +

# Add a title above the map, positioned to the left
tm_title(
    "Proximity to Chemical Facilities\nin East Baton Rouge Parish, LA",
    size = 1.0,
    position = tm_pos_out("center", "top", pos.h = "left", pos.v = "bottom"))

```

Proximity to Chemical Facilities in East Baton Rouge Parish, LA



Results

Based on these two maps, we can see there is a pattern between where people of color live and proximity to RMP facilities in East Baton Rouge Parish, Louisiana. According to the EPA, an RMP facility is any stationary industrial source that stores, handles, or processes extremely hazardous toxic substances (EPA, 2026). According to the maps, the block groups with the highest RMP proximity scores are concentrated in the northern part of East Baton Rouge Parish along the Mississippi River. Most of these communities are 80–100% people of color. When we look at the northeastern part of the parish, we can see that the proportion of people of color living there is less than 10%, and these areas are also the farthest from RMP facilities. It is important to note that the RMP variable used in this analysis only measures proximity to the facilities, not actual exposure, and these maps show a correlation rather than

a cause. If we were to take this a step further, we could examine the different health problems that these communities live with. In conclusion, these maps offer a good understanding of how the distribution of people of color in East Baton Rouge Parish varies spatially in relation to proximity to RMP chemical facilities, and they highlight where efforts to mitigate these social inequities could be focused.

Citation

Environmental Protection Agency. (2026). *Risk Management Program (RMP) Rule*. EPA. <https://www.epa.gov/rmp>